

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer **ALL** questions in this section. Marks are indicated at the end of each question. Together they are worth 80% of the total marks for this examination.

Question 1 (20 marks – approximately 36 minutes)

Fish-Water Farm is a leading aquaponics farming company founded locally in Hong Kong. The company is making use of its well-developed aquaponics eco-system, combined with technologies for vertical farming and a lighting system for photosynthesis, in growing organic foods such as vegetables, fruits, herbs, and edible flowers, amongst others.

The agricultural crops are fed with natural nutrients which are free from chemicals by managing a perfect balance between fishes and plants to operate the aquaponics eco-cycle. This starts with converting the wastes produced by fishes into fertiliser for plants and then using the plants to filter the water for the fish.

Apart from being the suppliers of local supermarkets, Fish-Water Farm also operates an online shop for selling its diversified selection of fishes, vegetables and fruits. One of its best-selling products is a 100g salad box with cherry tomatoes that sells at HK\$58 each. During the year 2020, the total annual expenditures for producing 36,000,000g vegetables for the salad box are extracted below:

	HK\$
Cost of building the new indoor aquaponics farm	20,000,000
Direct labour costs	2,950,000
Fish feed and seeds supplies	1,380,000
Water	2,780,000
Electricity	3,698,000
Transportation	738,000
Veterinary and hygiene	810,000
Fixed overheads	8,056,000
Depreciation	1,894,000
Sundry expenses	1,286,000

To develop a strong understanding of food safety and urban farming knowledge, Fish-Water Farm offers a variety of free green workshops and seminars to school children and adults. Free site visits to the company's indoor planting laboratory, one of Hong Kong's largest state-of-the-art indoor aquaponics farms, can also be booked in advance online.

Required:

- (a) Identify and explain **THREE** factors that affect the pricing policies of Fish-Water Farm.

(6 marks)
- (b) Calculate the unit cost (round to one decimal place) for each 100g salad box sold by Fish-Water Farm.

(2 marks)
- (c) Based on the answer calculated in Question 1(b), identify the pricing strategy adopted by Fish-Water Farm for charging a unit price of HK\$58 for each 100g salad box and explain the benefit of this pricing strategy.

(6 marks)
- (d) Explain the rationale for providing the free green workshops and site visit activities to the public. Describe any associated pricing strategies to promote the products or to increase sales during the free activities organised by Fish-Water Farm.

(6 marks)

Question 2 (20 marks – approximately 36 minutes)

Urban Park Restaurant ("Urban Park") owns and operates a German and Italian catering business serving middle-income customers. This business has been gaining popularity for one of its signature dishes, a mixed salad consisting of crispy pork knuckle with potato wedges. Urban Park has accumulated a sizeable amount of cash in recent years.

The management of Urban Park intends to expand its operations by acquiring a wine company or a hand-crafted beer company. As the Financial Controller of Urban Park, you are asked to evaluate the business valuation plan with the below given information of the two companies.

Wine company

The weighted average cost of capital of the business is now 10%. The projected free cash flows for Years 1 and 2 are HK\$1,500,000 and HK\$2,000,000, respectively. The projected financial information for Year 3 is extracted below:

	HK\$
Sales revenue	5,500,000
Operating expenses	620,000
Depreciation	215,000
Interest	96,000
Capital expenditures	600,000
Additions to working capital	250,000

Cost of goods sold is 20% of sales revenue. The company is in the tax bracket of 20%. After Year 3, free cash flow to assets is projected to grow at a constant rate of 6%.

Hand-crafted beer company

There are one million ordinary shares issued by the company. The earning per share and current market price per share of the company are HK\$5.83 and HK\$14.34, respectively, and the weighted average price-to-earnings ("P/E") ratio of all the stocks of comparable companies is 10.25.

Required:

- (a) **Given that a company's value should be equal to the present value of all the future free cash flows to assets, calculate free cash flow for Year 3 and thus the firm value of the wine company.**
(13 marks)
- (b) **Compute the firm value of the hand-crafted beer company using the P/E ratio approach.**
(3 marks)
- (c) **Apart from the financial evaluations above, advise on any other non-financial considerations for the investment appraisal.**
(4 marks)

Question 3 (20 marks – approximately 36 minutes)

Toppoint Corporation ("Toppoint") is a tablet computer manufacturer who is planning to expand its business by acquiring one of its electric parts suppliers. Vertex Company ("Vertex") and Peak Company ("Peak") are the two best performing companies in the industry. Their financial statements are listed as follows:

Statement of financial position as at 31 December 2020	<u>Vertex</u> HK\$	<u>Peak</u> HK\$
Non-current assets	4,459,283	5,321,092
Current assets		
Inventories	214,275	312,879
Accounts receivable	328,200	438,464
Cash	824,912	504,514
	<u>1,367,387</u>	<u>1,255,857</u>
Total asset	<u>5,826,670</u>	<u>6,576,949</u>
Long-term liabilities	2,641,425	1,462,369
Current liabilities	<u>705,210</u>	<u>828,239</u>
Total liabilities	<u>3,346,635</u>	<u>2,290,608</u>
Share capital*	1,024,760	3,129,993
Retained earnings	<u>1,455,275</u>	<u>1,156,348</u>
Total equity	<u>2,480,035</u>	<u>4,286,341</u>
Total liabilities and equity	<u>5,826,670</u>	<u>6,576,949</u>
*Number of shares	<u>40,000</u>	<u>30,000</u>

Statement of profit or loss for the year ended 31 December 2020

	HK\$	HK\$
Net sales (all credit)	7,054,748	7,786,691
Cost of goods sold	1,264,983	1,524,683
Operating expenses	4,422,273	5,099,828
Depreciation	<u>188,745</u>	<u>202,660</u>
Income before interest and tax	<u>1,178,747</u>	<u>959,520</u>
Interest	<u>301,040</u>	<u>281,622</u>
Taxable income	<u>877,707</u>	<u>677,898</u>
Income tax	<u>228,756</u>	<u>200,803</u>
Net income	<u>648,951</u>	<u>477,095</u>

Required:

- (a) Calculate the missing ratios for Vertex and Peak in the following table. Round your answers to two decimal places.

<u>Liquidity:</u>	<u>Vertex</u>	<u>Peak</u>
Current ratio	1.94	1.52
Quick ratio	?	1.14
<u>Asset management efficiency:</u>		
Inventory turnover	?	4.87
Days' receivables (assume 365 days in a year)	16.98	20.55
Total asset turnover	?	1.18
<u>Leverage:</u>		
Debt ratio	0.57	?
Equity multiplier	?	1.53
Times-interest-earned	?	3.41
<u>Profitability:</u>		
Net profit margin	?	6.13%
Return on assets	11.14%	?
Return on equity ("ROE")	?	?

(5 marks)

- (b) Analyse the financial performances of Vertex and Peak in the following aspects:
- i. Liquidity (1 mark)
 - ii. Asset management efficiency (2 marks)
 - iii. Leverage (3 marks)
 - iv. Profitability (2 marks)
- (c) Analyse the ROE of Vertex and Peak using the DuPont formula, and explain which company should have a better ROE and which one should be exposed to higher financial risks. (5 marks)
- (d) By comparing the overall financial performances of Vertex and Peak, advise on and explain which company Toppoint should acquire. (2 marks)

Question 4 (20 marks – approximately 36 minutes)

Good Taste Restaurant ("the Company") is considering a four-year renovation project to replace its old kitchen facilities and equipment. You are the Financial Manager of the Company and are responsible for estimating the cost of capital of the Company employed for capital budgeting evaluation. The capital structure of the Company is listed below:

	<u>No. of stocks/ bonds</u>
Bonds (7% semi-annual coupon, HK\$1,000 face value)	13,000
Preferred stock (issued at HK\$100 per share, 6% dividend)	25,000
Common stock (issued at HK\$10 per share)	120,000

Five years ago, the Company issued the 15-year, 7% semi-annual coupon bonds that are still outstanding now and are currently selling at HK\$1,000 each. Market prices per share for the preferred stock and the common stock are HK\$98 and HK\$52, respectively. The management of the Company just paid an annual dividend of HK\$1.4 per common stock, and the market expects that all future dividends will constantly grow by 5% per annum for the foreseeable future. The common stock has a beta coefficient of 2.3. The treasury bill currently yields 4%, and the expected return on the market is 15%. The Company is in the 40% corporate tax bracket.

Detailed information of the initial cash outflows and cash inflows of the two four-year renovation projects are listed as follows:

Year	Project A (HK\$)	Project B (HK\$)
0	-1,680,000	-730,000
1	200,000	230,000
2	350,000	350,000
3	570,000	400,000
4	820,000	510,000

Required:

(a) Calculate the following:

- (i) cost of common stock (R_E) by taking the average of using both the dividend growth model and the capital asset pricing model;
- (ii) cost of preferred stock (R_P); and
- (iii) pre-tax cost of debt (R_D).

Present your answers as a percentage with two decimal places.

(8 marks)

- (b) Assume that the Company is going to maintain the current capital structure. Calculate the weighted average cost of capital ("WACC") of the Company.

Present your answers as a percentage with two decimal places.

(5 marks)

- (c) Compute the net present values for both Project A and Project B, and identify which project the Company should choose.

(5 marks)

- (d) Given the answers computed in Question 4(c), explain whether the internal rate of return should be larger or smaller than the WACC computed in Question 4(b) for both Project A and Project B.

(2 marks)

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